

Def 5.27 (Quantum circuit)

Let $n, L \in \mathbb{N}$ and $U_1, \dots, U_L \in \mathcal{U}(\mathbb{H}^{\otimes n})$ be quantum gates

$U = U_L \dots U_1$; plain quantum circuit

$\in \mathcal{U}(\mathbb{H}^{\otimes n})$

L : (length / depth) of quantum circuit.

$$\rho \in \mathcal{D}(\mathbb{H}) \longrightarrow \boxed{U} \longrightarrow U\rho U^* \in \mathcal{D}(\mathbb{H})$$

$$|\psi\rangle \in \mathbb{H}, |\psi\rangle\langle\psi| \xrightarrow{U} U|\psi\rangle\langle\psi|U^* = |U\psi\rangle\langle U\psi|$$

$U|\psi\rangle$

WTD: For given $f: \mathcal{N}_0 \longrightarrow \mathcal{N}_0$, construct quantum circuit which implies the value of f .

Def 5.31

① (factor-wise binary addition)

$$\boxplus: \mathbb{H}^{\otimes n} \otimes \mathbb{H}^{\otimes n} \longrightarrow \mathbb{H}^{\otimes n}$$

$$|a\rangle \otimes |b\rangle \longmapsto |a\rangle \boxplus |b\rangle := \bigotimes_{j=1}^n |a_j \oplus b_j\rangle$$

② Let $f: \mathcal{N}_0 \longrightarrow \mathcal{N}_0$, $n, m \in \mathbb{N}$, $\mathbb{H}^A := \mathbb{H}^{\otimes n}$, $\mathbb{H}^B := \mathbb{H}^{\otimes m}$

We say a circuit U_f implements f on $\mathbb{H}^A \otimes \mathbb{H}^B$

$$\Leftrightarrow U_f: \mathbb{H}^A \otimes \mathbb{H}^B \longrightarrow \mathbb{H}^A \otimes \mathbb{H}^B$$

$$|x\rangle \otimes |y\rangle \longmapsto |x\rangle \otimes |y \boxplus f(x)\rangle \quad) \textcircled{1}$$

Remark ① Why the definition Def 5.31 ② good??

$$U_f(\mathbb{H}^{\otimes n} |0\rangle^n \otimes |0\rangle^m) \quad \mathbb{H}|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$= U_f\left(\underbrace{\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \dots \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}}_{n\text{-times}} \otimes |0\rangle^m\right)$$

$$= U_f\left(\frac{1}{2^{\frac{n}{2}}} \sum_{x=0}^{2^n-1} |x\rangle^n \otimes |0\rangle^m\right) \stackrel{\textcircled{1}}{=} \frac{1}{2^{\frac{n}{2}}} \sum_{x=0}^{2^n-1} |x\rangle \otimes |f(x)\rangle$$

② U_f is unitary.

$\{|x\rangle \otimes |y\rangle\}$ be ONB of $H^A \otimes H^B$.

$$|\Psi\rangle \in H^A \otimes H^B, \quad |\Psi\rangle = \sum_{x=0}^{2^n-1} \sum_{y=0}^{2^m-1} \Psi_{xy} |x\rangle \otimes |y\rangle$$

$$\begin{aligned} \|U_f \Psi\|^2 &= \langle U\Psi | U\Psi \rangle = \sum_{x,y,a,b} \overline{\Psi_{x,y}} \Psi_{a,b} \underbrace{\langle x|a \rangle}_{\delta_{xa}} \underbrace{\langle y|b \oplus f(x)\rangle}_{\delta_{yb}} \\ &= \|\Psi\|^2 \end{aligned}$$

$\Rightarrow U_f$ is unitary

Thm 5.28 Let $H^{I/O}$ and H^W be Hilbert space and

$|w_i\rangle, |w_f\rangle \in H^W$ s.t. $\|w_i\| = \|w_f\| = 1$.

And let $\hat{U} \in \mathcal{U}(H^{I/O} \otimes H^W)$ s.t.

for any $|\Psi\rangle \in H^{I/O}$, $\hat{U} |\Psi \otimes w_i\rangle = (U|\Psi\rangle) \otimes |w_f\rangle$

Then, we have

① $U \in \mathcal{U}(H^{I/O})$

② $\text{tr}^W(\hat{U}(\rho \otimes \rho_{w_i})\hat{U}^*) = U\rho U^*$

$\rho \in \mathcal{D}(H^{I/O})$

$\rho_{w_i} = |w_i\rangle\langle w_i|$

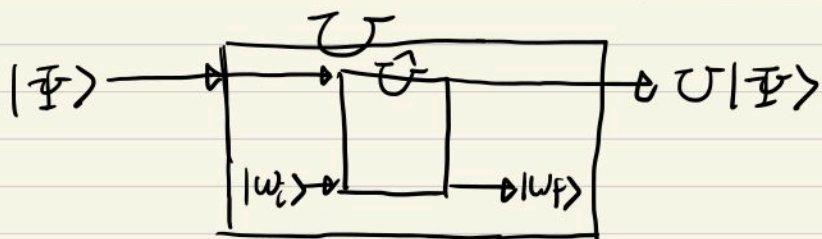
Def 5.29 (Quantum circuit with ancilla)

Let $H^{I/O} = \otimes^n H$, $H^W = \otimes^m H$.

Circuit U on $H^{I/O}$ is said to be "circuit with ancilla" implemented with ancilla $|w_i\rangle \in H^W$

$\Leftrightarrow \exists |w_i\rangle, |w_f\rangle \in H^W, \exists \hat{U} \in \mathcal{U}(\otimes^{n+m} H)$ s.t.

for any $|\Psi\rangle \in H^{I/O}$, $\hat{U} |\Psi \otimes w_i\rangle = (U|\Psi\rangle) \otimes |w_f\rangle$



pf of Thm 5.28

① U is unitary

Fix $|\Psi\rangle \in H^{I_0}$.

$$\begin{aligned}\|U\Psi\|^2 &= \langle U\Psi | U\Psi \rangle \cdot \langle \omega_f | \omega_f \rangle \\ &= \langle U\Psi \otimes \omega_f | U\Psi \otimes \omega_f \rangle \\ &= \langle \hat{U}|\Psi \otimes \omega_i\rangle | \hat{U}|\Psi \otimes \omega_i\rangle \rangle \\ &= \|\hat{U}|\Psi \otimes \omega_i\rangle\|^2 = \|\Psi \otimes \omega_i\|^2 \\ &= \|\Psi\|^2 \|\omega_i\|^2 = \|\Psi\|^2\end{aligned}$$

② $\text{tr}^W(\hat{U}(\rho \otimes \rho_{\omega_i})\hat{U}^*) = \text{tr}^W(\rho)$

Let $\rho = \sum_j \rho_j |\Psi_j\rangle \langle \Psi_j|$ for $\{|\Psi_j\rangle\}$ ONB of H^{I_0}

$$\begin{aligned}\hat{U}(\rho \otimes \rho_{\omega_i})\hat{U}^* &= \hat{U}(\sum_j \rho_j |\Psi_j\rangle \langle \Psi_j| \otimes |\omega_i\rangle \langle \omega_i|)\hat{U}^* \\ &= \sum_j \rho_j \hat{U}|\Psi_j \otimes \omega_i\rangle \langle \Psi_j \otimes \omega_i| \hat{U}^*\end{aligned}$$

$$\begin{aligned}\langle \Psi_j \otimes \omega_i | \hat{U}^* &= \langle \hat{U}(\Psi_j \otimes \omega_i) | \\ &= \langle (U\Psi_j) \otimes \omega_f | = \langle U\Psi_j | \otimes \langle \omega_f | \end{aligned}$$

$$\begin{aligned}\hat{U}(\rho \otimes \rho_{\omega_i})\hat{U}^* &= \sum_j \rho_j (|U\Psi_j\rangle \otimes |\omega_f\rangle)(\langle U\Psi_j| \otimes \langle \omega_f|) \\ &= \sum_j \rho_j \underbrace{|U\Psi_j\rangle \langle U\Psi_j| \otimes |\omega_f\rangle \langle \omega_f|}_{= U|\Psi_j\rangle \langle \Psi_j| U^*}\end{aligned}$$

$$\underbrace{\sum_j \rho_j U|\Psi_j\rangle \langle \Psi_j| U^*}_{= U\rho U^*} \otimes |\omega_f\rangle \langle \omega_f|$$

operator on H^{I_0}

$$\text{tr}^W(\hat{U}(\rho \otimes \rho_{\omega_i})\hat{U}^*) = \text{tr}^W(\underbrace{U\rho U^*}_{\text{operator on } H^{I_0}} \otimes \underbrace{|\omega_f\rangle \langle \omega_f|}_{\text{operator on } H^W})$$

$$= U_\rho U^* \cdot \text{tr}(|\psi\rangle\langle\psi|) = U_\rho U^* \\ (\because \|\psi\| = 1)$$

Def 5.32 $U_{\boxplus} : \mathbb{H}^{\otimes m} \otimes \mathbb{H}^{\otimes m} \longrightarrow \mathbb{H}^{\otimes m} \otimes \mathbb{H}^{\otimes m}$

$$|a\rangle \otimes |b\rangle \longmapsto |a\rangle \otimes |a \boxplus b\rangle$$

Lemma 5.33 $U_{\boxplus}$ is unitary

pf) ① $U_{\boxplus}^* = U_{\boxplus}$

$$\langle U_{\boxplus} |a\rangle \otimes |b\rangle | |c\rangle \otimes |d\rangle \rangle = \langle |a\rangle \otimes |a \boxplus b\rangle | |c\rangle \otimes |d\rangle \rangle \\ = \langle a | c \rangle \langle a \boxplus b | d \rangle \dots (*)$$

$$\langle |a\rangle \otimes |b\rangle | U_{\boxplus} |c\rangle \otimes |d\rangle \rangle = \langle a | c \rangle \langle b | c \boxplus d \rangle \dots (**)$$

i) $a \neq c$

$$(*) = (**) = 0$$

ii) $a = c$

$$(*) = \langle a \boxplus b | d \rangle = \langle b | a \boxplus d \rangle = (**)$$

② $U_{\boxplus}^2 = \mathbb{I}$

$$U_{\boxplus}^{-1} (|a\rangle \otimes |b\rangle) = U_{\boxplus} (|a\rangle \otimes |a \boxplus b\rangle) \\ = |a\rangle \otimes |a \boxplus (a \boxplus b)\rangle = |a\rangle \otimes |b\rangle$$

$\therefore$ By ① & ② $U_{\boxplus}$ is unitary.

Thm 5.34 Let $f: \mathcal{N}_0 \rightarrow \mathcal{N}_0$ and $n, m \in \mathbb{N}$.
 $H^A := {}^q H^{\otimes n}$, $H^B := {}^q H^{\otimes m}$.

Suppose there exist A_f and B_f which are circuit on $H^A \otimes H^B$ s.t. there exist $|w_i\rangle, |w_f\rangle \in H^B$ satisfying

$$\begin{cases} A_f(|x\rangle \otimes |w_i\rangle) = |\psi(x)\rangle \otimes |f(x)\rangle \text{ for some } \psi \\ B_f(|\psi(x)\rangle \otimes |f(x)\rangle) = |x\rangle \otimes |w_f\rangle \end{cases}$$

Then, $\hat{U}_f$ defined by (on $H^A \otimes H^B \otimes H^B$)

$$\hat{U}_f := (\mathbb{1}_A \otimes S^{B,B}) (B_f \otimes \mathbb{1}_B) (\mathbb{1}_A \otimes U_{\boxplus}) (A_f \otimes \mathbb{1}_B) (\mathbb{1}_A \otimes S^{B,B})$$

where $S^{B,B}: |b_1\rangle \otimes |b_2\rangle \mapsto |b_2\rangle \otimes |b_1\rangle$; unitary

satisfies

$$\hat{U}_f(|x\rangle \otimes |y\rangle \otimes |w_i\rangle) = |x\rangle \otimes |y \boxplus f(x)\rangle \otimes |w_f\rangle$$

Remark ① If we get $\hat{U}_f$, then from the def. of quantum circuit with ancilla we can obtain

$$U_f(|x\rangle \otimes |y\rangle) = |x\rangle \otimes |y \boxplus f(x)\rangle$$

② How to get A and B ?? (on sec 5.5)

pf of Thm 5.34

$$\begin{aligned}
 & \hat{U}_f (|x\rangle \otimes |y\rangle \otimes |w_i\rangle) \\
 &= (\mathbb{1}_A \otimes S^{B,B}) (B_f \otimes \mathbb{1}_B) (\mathbb{1}_A \otimes U_{\oplus}) (A_f \otimes \mathbb{1}_B) (\mathbb{1}_A \otimes S^{B,B}) (|x\rangle \otimes |y\rangle \otimes |w_i\rangle) \\
 & \quad \underbrace{\hspace{15em}}_{= |x\rangle \otimes |w_i\rangle \otimes |y\rangle} \\
 & \quad \underbrace{\hspace{10em}}_{= |\psi(x)\rangle \otimes |f(x)\rangle \otimes |y\rangle} \\
 & \quad \underbrace{\hspace{10em}}_{= |\psi(x)\rangle \otimes |f(x)\rangle \otimes |f(x) \oplus y\rangle} \\
 & \quad \underbrace{\hspace{15em}}_{= |x\rangle \otimes |w_f\rangle \otimes |y \oplus f(x)\rangle} \\
 &= |x\rangle \otimes |y \oplus f(x)\rangle \otimes |w_f\rangle
 \end{aligned}$$